ENEL 585 Final Project

Fall 2026 | Common requirements

Oct. 30, 2026 · Common requirements for Projects A–E

Design and experimentally validate a controller for your chosen converter. Equipment is provided; your group assembles and verifies external wiring, establishes operating limits, designs and tests the controller in simulation and HIL, and compares predictions with bench measurements.

Your station sheet defines your converter, its hardware limits and the bench tests; read it with this handout, the Inverter Box Setup Guide and the PLECS Setup Guide.

How to read this handout. Section 1 summarizes each deliverable; Section 2 shows how to plan your three bench sessions. Each deliverable's full requirements and rubric are in its own appendix (A–E), followed by reference material (F–I).

Changes since the Oct. 1 version: the design review moves to the start of Bench 2 (Bench 1 ungraded) and includes a switched simulation of one station-sheet transient; HIL sign-off is planned for Bench 2.

1 Deliverables

Deliverable	Course weight	Due or scheduled time	Description
Proposal (App. A)	4%	Fri Oct 9	Your proposal fixed your project and the operating points, targets and protection thresholds your later work is assessed against. A later change is re-reviewed before you energize and carries no penalty when justified; report the original and revised values with the reason.
Oral design review (App. B)	8%	Bench 2: Nov 20 (B01), Nov 27 (B02); optionally Bench 1	A ten-minute oral review in which the whole group defends its operating points, switching frequency, control design, sensing, protection and bring-up. A group that is ready may take it at Bench 1 instead.
HIL sign-off (App. C)	Ungraded; required	Start of Bench 2; Bench 3 at the latest	Demonstrate your controller on the remote HIL model to the instructor or TA. Required before any closed-loop hardware operation. Signing off at Bench 3 instead of Bench 2 leaves one session for closed-loop operation and all measurements.
Final presentation (App. D)	10% group + 8% individual questions	Registrar-scheduled examination slot	The group presents its design and bench evidence; each student then answers individual questions.
Final report (App. E)	10%	Fri Dec 11	About 10 pages comparing your predictions with bench measurements and relevant PLECS and Typhoon files.

Total: 40% of the course grade.

2 Planning your three bench sessions

Session	Goal for the session
1 Bring-up Oct 30 (B01) Nov 6 (B02)	Wire your station and check the wiring against your connection diagram. Verify your startup and shutdown procedure, calibrate your probes and check your gating signals. If time allows, start open-loop tests at reduced power.
2 Closed loop Nov 20 (B01) Nov 27 (B02)	Design review and HIL sign-off at the start. Then run your first closed-loop tests at reduced power, increasing power in steps.
3 Measurements Dec 4 (B01) Dec 1 (B02), 10:00–12:15	Finish closed-loop testing and capture every operating point and station-sheet test for the presentation and report. A group that did not sign off on HIL at Bench 2 does so at the start of this session.

Suggestions:

- **Use remote HIL before Bench 2.** Bench time is short, and without HIL testing you are unlikely to finish closed-loop testing in time. Closed-loop hardware operation needs HIL sign-off; the start of Bench 3 is the latest.
- **Keep a bench log** of conditions, probe scaling and model and firmware versions for every capture. The report ZIP must include it.
- **If you share a station,** agree with the other group how you will leave the wiring, and record the configuration you leave.
- **Before you energize at Bench 1,** the instructor or TA must sign off your wiring, probe scaling, gate mapping, protection settings and startup procedure, including precharge for C–E.

Appendix A Proposal

A.1 Proposal requirements (issued Oct. 1; timing and the ITECH limit updated)

Two pages plus figures. The proposal fixes your project and sets the operating points and targets your later work is assessed against. No controller design is required; desired crossover and phase margin, control structure and compensator design are presented at the design review. Simulation is encouraged but not required; if you include simulation results, state what the model includes and assumes. Include:

1. Operating envelope and points. The envelope is the range of voltage, current and power (E: reactive power) your station can deliver without exceeding any hardware limit on the station sheet or in the component datasheets. The *binding constraint* is the limit you reach first as you raise power, for example a switch or inductor current rating, the DC voltage limit, the source limit (the ITECH 10 A current limit for A/B; about 19 A per transformer winding and the 16 A bypass breaker for C–E) or the load bank. Choose at least two operating points inside the envelope (E: one inductive and one capacitive command) and state each point's *margin* to the relevant limits; for example, 15 A peak against a 20 A rating is a 25% margin. Margin covers ripple, component tolerance, measurement error and overshoot during load changes.

2. Steady-state targets. For each operating point, give measurable values you will check on the bench. Calculate those set by the circuit and operating point, such as voltage, current and power. Targets that depend on how well your controller performs, such as input-current THD and power factor for C and D, are taken from a cited standard, application note or comparable published design, with reasons they suit your station. Targets may be revised later with justification and without penalty.

3. Preliminary switching frequency. Choose a frequency at or below 10 kHz and give your reasons, for example the trade-off between ripple and switching losses, or the harmonic spacing you observed in Lab 2. An estimate of ripple or peak current is welcome but optional: over one switching period the line voltage is nearly constant, so volt-second balance can be applied at the worst-case instant. The frequency is revisited at the design review.

4. Sensing and protection. List each signal you will measure, with its probe, probe setting and ADC scaling (the voltage at the ADC pin at the largest expected value), keeping your trip thresholds inside the ADC range. State your protection thresholds, the overcurrent and overvoltage levels set above your highest operating point and below the hardware limit, and what your firmware does when they trip.

5. Connection diagram. Extend your station-sheet figure, which shows the power circuit only, into the wiring you will build: add each probe and its connection points, the ADC input it feeds, the PWM output driving each switch, the load connections and the instruments.

6. Safety plan. Using Appendix H, write your station's procedures as numbered steps:

- **Startup:** the order of supplies, breakers, load and PWM enable.
- **Normal shutdown:** the reverse sequence, ending with the DC link confirmed below 5 V.
- **Fault response:** what you do if a protection threshold trips or the converter behaves unexpectedly: the control that removes power (ITECH output for A/B, B2 for C–E) and how you confirm the DC link is below 5 V before handling.

7. Work plan. A table of tasks, each with a lead, a supporting member and a target date, through Bench 3. Half a page is enough. Include at least:

- **Before Bench 1: your switched PLECS model of the station and your connection diagram finalized. HIL checks (model parameters, I/O mapping and first controller communication) begin once your project's model is posted (Appendix C); Bench 1 does not depend on them.**
- **Design review:** small-signal plant model, desired crossover and phase margin, compensator and a switched simulation of one station-sheet transient, ready by the start of Bench 2 (Bench 1 if you take the review early).
- **Before Bench 2:** firmware and the HIL checklist.
- **Bench 3:** the measurements and who takes them.

- **Report:** who drafts each section.

Plan around the lecture schedule: controller design depends on the dynamics and control lectures, and three-phase work (A) on the three-phase lectures.

8. AI disclosure. Name any AI tools you used and what you used them for, or state that none were used (Appendix I).

A.2 Proposal rubric: 4 course points

Criterion	Weight	Full-marks standard
Hardware analysis and operating points	1.5	Derives the operating envelope from the station sheet's hardware limits, identifies the binding constraint, and chooses operating points with stated margins.
Steady-state targets and preliminary switching frequency	1	States measurable steady-state targets for each operating point; calculates those set by the circuit and operating point, and justifies controller-dependent targets from a cited source adapted to the station (the justification is assessed, not the value). Gives a reasoned preliminary frequency at or below 10 kHz; a ripple or peak-current estimate is optional.
Sensing and safety plan	1	Specifies the measured signals, probe settings, ADC scaling and protection thresholds; gives correct station-specific startup, shutdown and fault-response steps, including precharge for C–E.
Work plan	0.5	Gives a task table with leads, support and dates through Bench 3, covering HIL checks, the design review, firmware, measurements and the report, planned around the lecture schedule.

Appendix B Design review

B.1 Requirements

Format. A ten-minute oral review at the start of Bench 2, before closed-loop hardware operation. Spend about four minutes walking through your design using your calculations and PLECS models on paper or screen; slides are not needed. The instructor or TA may ask questions as you go, and the remaining time is questions. All members attend, and any member may be asked about any part of the design. Answers must be unaided by AI.

Early option. A group that is ready may take the review at Bench 1 by emailing the instructor at least three days before. The early review is your one graded review, with no second attempt at Bench 2.

What to cover. Your operating points, switching frequency and protection settings from the proposal, with any refinements and the reason for them. Your control design: control objectives and structure, plant model, compensator, and desired and predicted crossover and phase margin. Show a switched PLECS simulation of at least one station-sheet transient and compare it with your prediction. HIL results are welcome but not required.

B.2 Rubric: 8 course points

Criterion	Weight	Full-marks standard
Switching frequency and operating limits	2	Defends the operating points and a switching frequency at or below 10 kHz with reasons: ratings, losses, bandwidth or resonance.
Sensing and protection	1.5	Explains the ADC scaling verified on hardware, the PWM-to-switch mapping and the protection settings, consistent with the design.
Bring-up readiness	1.5	Explains the connection diagram, how the wiring was verified, and startup and precharge. Both groups sharing a station are expected to explain it equally well.
Control design	3	States the desired crossover and phase margin; derives the plant and designs the compensator systematically; predicts the achieved crossover and phase margin; the switched simulation agrees with the prediction or the difference is explained.

All members must be able to explain the work. The review is group-marked; individual understanding is assessed separately at the final presentation (Appendix D).

Appendix C HIL sign-off (ungraded, required)

Models. Remote HIL access instructions are on D2L. Plant models are posted on D2L project by project. They reproduce the probe and interface-board sensing (1.62 V ADC offset, Inverter Box Setup Guide v3), so the same firmware scaling works on the HIL and the bench. Once your project's model is posted, check model parameters and I/O mapping and attempt controller communication. Bench 1 does not depend on HIL.

Checklist, complete before your sign-off. Startup, stable operating points, every station-sheet transient, C–E synchronization lock, relevant limits and protective shutdown. Further tests are optional.

Sign-off. Before any closed-loop hardware operation, demonstrate the checklist on HIL to the instructor or TA, identify versions and settings, and explain the response. Plan to sign off at the start of Bench 2; earlier sign-off is welcome. The start of Bench 3 is the latest, and leaves that one session for closed-loop operation and all measurements. Staff record the sign-off; there is no separate submission or advance deadline. Repeat failed checks before progressing, recheck after material changes, and keep the evidence for the report and demonstration.

HIL does not replace checks of physical wiring, sensor polarity and hardware protection.

Appendix D Final presentation

D.1 Requirements

Group demonstration. In the registrar-scheduled examination slot, present your design and the evidence collected at the bench sessions; no hardware is operated. Show your reviewed targets, predicted and measured performance, relevant HIL results, and the conclusions the comparison supports. All members contribute. The format, room, group schedule and time allocation will be posted on D2L at least two weeks beforehand.

Individual questions. Each student answers unaided questions about a design decision and a measured result, with follow-ups as needed; questions may draw on any part of the project. Students receive comparable opportunities, and exhaustive coverage is not expected. This mark (8% of the course) is excluded from peer adjustment.

D.2 Group demonstration rubric: 10 course points

Criterion	Weight	Full-marks standard
Experimental evidence	4	Presents evidence of safe, stable closed-loop operation at each operating point and every station-sheet test, captured at the bench sessions. States conditions and signals; compares with targets and required HIL evidence, and quantifies shortfalls. Additional bench transient tests are not required for full marks.
Comparison and explanation	4	Connects predictions to measurements; supports explanations with evidence and identifies unresolved causes and useful further tests.
Communication	2	Presents clear, scaled figures and a coherent technical account within the allotted time; every member contributes.

D.3 Individual questions rubric: 8 course points

Understanding of the design — 8 points. Explains a design decision and a measured result with correct reasoning, including follow-up questions. Assessed individually using the common quality scale, with comparable opportunities to demonstrate understanding.

Appendix E Final report

E.1 Requirements

About 10 pages, excluding appendices, written for a reader who knows power electronics but has not seen your station. Organize it around:

- Station, operating points and targets, and system diagram.
- Hardware analysis, switching-frequency choice and control design.
- Simulation and remote HIL validation, including important model limitations.
- Experimental results and a compact table comparing required performance quantities with predictions and targets.
- Interpretation, main loss estimates, efficiency comparison if measured, unresolved questions and proposed improvements.

Label plots with axes, units, operating conditions and probe scaling, and identify the model and firmware versions used for the reported tests. Include an AI disclosure statement. Submit the PDF with a supporting ZIP of the PLECS models, controller export and settings, relevant HIL model changes, raw data and bench log; put calculations and supporting figures in report appendices.

E.2 Rubric: 10 course points

Criterion	Weight	Full-marks standard
Hardware analysis	2	Establishes operating limits, current margins and sensing requirements and uses them in design decisions. Explains AC-filter role and relevant resonance; derives ripple only where needed for the topology.
Control and model validation	3	Presents models, compensators and predicted responses; C–E show synchronization lock and explain how the provided PLL model is used; compares linearised, switched and HIL results and identifies model limitations.
Experimental evidence and comparison	3	Provides the station-sheet waveforms and metrics at each operating point, with conditions and scaling, compared against predictions and targets. Includes every station-sheet transient. Additional bench operating points or transients are not required for full marks.
Interpretation and losses	2	Uses evidence to discuss discrepancies and uncertainty; estimates main losses, explains operating/thermal implications and proposes justified improvements. Compares efficiency if measured.

Appendix F Project scope and what is provided

Groups of up to four, within a lab section, each chose one of Projects A–E; projects may repeat, and A is limited to two groups per section. Choices were fixed at proposal submission. Groups sharing a station across sections must each verify its wiring. B–E normally use single-phase banks; a three-phase bank may be assigned if needed. Use the resistance settings and connections issued for your assigned bank.

Station-sheet figures omit sensing and gate drives; C–E figures show the breaker and precharge paths. Crossings without dots are not connections.

Provided	Your responsibility
Converter equipment, instruments and controller hardware	Connection diagram, external wiring and verification, operating targets and protection settings
HIL plant models (power stage, sensing and I/O setup; C–E transformer voltage and precharge), with no control; remote access. C–E: example PLECS synchronization (PLL) model	Switched PLECS model of your station, controller design, firmware generated with PLECS Coder, HIL model configuration and remote validation
Station sheet: circuit, equipment and hardware limits	Operating points, switching frequency, bench measurements and interpretation

Appendix G Engineering requirements

Student wiring and shared stations

Each group prepares its own connection diagram. The first group assembles the external wiring; the second traces and verifies it. Make only justified, approved changes and record the configuration you leave. Both groups must explain and verify the station and obtain approval before energizing.

Operating limits and targets

Establish limits, probe settings, ADC scaling and protection from the station sheet, Setup Guide and component datasheets, including IGBT rating conditions. Explain the AC-filter role, tolerances and resonance. Compare simulated and measured peak/RMS filter currents, including ripple, with ratings. A separate AC-filter switching-ripple derivation is not required; calculate ripple where it supports a design choice, such as boost-inductor peak current or single-phase DC-link ripple.

Choose at least two feasible operating points (E: your inductive and capacitive reactive-power commands) with measurable targets. State them and your protection thresholds in the proposal; they are reviewed before you energize. Any later change is re-reviewed and carries no penalty when justified; report the original and revised values with the reason.

Switching frequency and controller design

Choose and justify a switching frequency at or below the 10 kHz lab ceiling, imposed by losses in the teaching inverter. Address losses, bandwidth and relevant resonance; include ripple/peak-current calculations only where needed for the design.

For each loop, derive the small-signal plant, choose the desired crossover and phase margin, design the compensator and predict the achieved values. State the sampling and PWM rates.

C–E synchronize to the 60 Hz transformer using the provided example PLECS model of a synchronization (PLL) block, which outputs the transformer-voltage angle; include it in your controller. C/D build their sinusoidal current reference from that angle, not from a scaled copy of the measured voltage; E sets its current phase from it for its P/Q commands. Show lock in simulation and HIL. A/B are standalone and need no synchronization. Each project has two differential voltage probes and two current probes. Read A/B DC input from the ITECH display.

Simulation and implementation

Compare switched and linearised PLECS models where the small-signal approximation applies. Generate LAUNCHXL-F28P55X firmware using PLECS Coder; preserve control logic and document settings through simulation, HIL and bench testing.

Your diagram must show power, gating, sensing/scaling, control loops, protection and instruments, including physical-to-ADC and PWM-to-switch mapping.

Measurements and interpretation

Demonstrate safe, stable closed-loop operation at each operating point and every test, signal and metric on your station sheet. Compare with predictions and explain limitations. Additional bench transients and efficiency measurements are optional. Define each metric, including how periodic ripple is treated. For C/D, phase error is the angle between the fundamentals of the transformer voltage and input current; distinguish it from true input power factor, and state the THD harmonic range and measurement bandwidth.

Estimate the main losses from datasheets to justify the operating points and switching frequency. Explain discrepancies between simulation and experiment using evidence.

Appendix H Bench procedures and safety

Why precharge is required

Diodes rectify AC with all gates off, so closing the supply onto a discharged DC link through inductance can overshoot: an ideal undamped series LC circuit, initially unenergized, charges its capacitor to twice an applied DC step. Actual peaks depend on source, circuit and damping; do not rely on supply OVP. On C–E, B_2 applies AC through R_{pc} with B_{bp} open; close B_{bp} once the DC link has settled. B_1 (16 A primary) de-energizes the transformer.

Connect the ITECH only to approved A/B DC-bus terminals, never C–E or a node that can exceed its voltage limit. Filter energy can return during transients; use the approved mode, limits and shutdown procedure.

Be ready to answer: With gating disabled, where does charging current flow, and what changes when the precharge bypass breaker closes?

Bench checks

Before energizing, obtain instructor/TA confirmation of wiring, probe scaling, gate mapping and protection. Independently verify shared wiring against your diagram. Measure below 5 V on the DC link and external capacitors before rewiring; switch auxiliary power off before handling gate BNCs. Interpret error inputs and LEDs using the Setup Guide sections Protection Related and Status Check.

Common startup and shutdown

Standard single-phase banks have 20 switched 250 Ω parallel branches (R = 250/n Ω for n closed switches); use the issued configuration and resistance settings for an assigned three-phase bank. The hardware limits on your station sheet are equipment ceilings, not operating targets. Commanded DC voltage is limited to 60 V, except C/D at 70 V. Your operating points and trip thresholds must keep peak and RMS stresses, including ripple and transients, within the component ratings; they are reviewed with your protection settings.

Use the sequence for your station below with your reviewed settings. Begin at a reduced-power starting point and increase in steps. Use only the intended bank switches for live load changes; select a reproducible change within your reviewed limits. On C–E, the instructor or TA runs your first precharge and energization with you and confirms the station controls and expected settled DC-link voltage.

ITECH supply for A and B

1. With PWM inhibited and the AC load connected, set the ITECH to source mode, output off and 0 V, with the current limit at 10 A and overvoltage protection at 60 V. Leave the station transformer (used by C–E) disconnected from the inverter and load.

2. After the instructor or TA approves your setup, enable the source and raise DC voltage in steps, to at most 58 V, while monitoring voltage and charging current. Enable PWM only after charging has settled, then approach your first operating point.

3. For normal shutdown, reduce demand, inhibit PWM, ramp the ITECH to zero and switch its output off. Ramping down pulls the DC link down with it; confirm the ITECH display reads below 5 V before handling. The filter capacitors discharge through their bleeders.

Transformer supply for C D and E

1. Identify the 16 A primary breaker B_1, the 20 A secondary breaker B_2 and the 16 A bypass breaker B_{bp}. With B_2 open and the DC link below 5 V, open all DC-load switches, inhibit PWM and open B_{bp}. Confirm the charging loop: 60 Ω through two resistors for C/D; one 30 Ω resistor with neutral return for E.

2. Close B_2 to apply AC through R_{pc}. Watch the DC-link voltage rise and settle near the expected unloaded value, roughly the input peak less diode drops. If it fails to rise, stalls or behaves abnormally, open B_2 and involve the instructor or TA.

3. Close B_{bp} before connecting load or enabling PWM; never operate the converter through R_{pc}.

4. Connect the starting load and enable switching in your reviewed order. C/D require a connected load before sustained boost operation; E may regulate with its switched test load open.

5. For normal shutdown, reduce current demand, inhibit PWM and open B_2. PWM inhibition does not stop passive diode charging, and opening or tripping B_{bp} leaves R_{pc} in circuit without isolating the converter. Measure below 5 V on the DC link and external capacitors before handling; open B_{bp} before restarting.

Faults and return to service

On a fault or unexpected behaviour, inhibit PWM and isolate the source (ITECH output off; B_2 open on C–E). Investigate with the instructor/TA before reset; do not repeatedly reset a trip. Recheck changed wiring or protection settings before energizing. Leave the station discharged and record its condition.

Appendix I Assessment and course policies

How marks are assigned

The rubrics in Appendices A, B, D and E give the full-marks standard for each criterion. Weights are course percentage points. Apply the scale below separately to each criterion; for example, 80% on a two-point criterion earns 1.6 course points.

Level	Band	Description
Excellent	90–100%	Meets the standard with clear, supported reasoning.
Good	75–89%	Meets most of the standard; minor gaps do not change the conclusions.
Adequate	60–74%	Essential content is present, but reasoning or evidence is limited.
Marginal	50–59%	Significant omissions or errors; the criterion is only partly addressed.
Inadequate	Below 50%	Little usable evidence, or errors that undermine the work.

The marker selects a level and a mark within its band, with a reason, and deducts a weakness under its own criterion; missing data does not exempt dependent analysis. Formatting matters only when it affects interpretation. Supported reasoning can earn full marks even if available evidence cannot resolve a discrepancy.

Group and individual contribution

The proposal, review, group demonstration and report receive group marks. Individual questions receive a separate mark for each student. Every student must understand the complete design, even where implementation work is divided.

Submit one proposal/report PDF per group to the D2L group dropbox by 23:59 on the due date, naming all members and the station. Each student submits confidential peer feedback separately through an individual D2L dropbox. Its form, deadline and instructions will be issued later, sufficiently before submission.

The instructor may adjust an individual student's group-marked subtotal by up to ±4 course points (±10 percentage points of the project), based on justified peer feedback, bench observations and that student's individual answers. The default is zero adjustment. The adjusted subtotal is bounded to 0–32, then the unchanged individual-question mark is added. Example: 24/32 group + 6/8 individual = 30/40; a +4 adjustment for that student gives 34/40.

Use of AI

AI tools are permitted for written work, models, code and study, with disclosure. Name the tools and describe their use in each submission. You are responsible for the correctness of all submitted work. The design review and individual questions assess your own understanding and must be answered without AI assistance. Undisclosed AI use is addressed under the course outline.

Absences and late work

Bench sessions cannot be replaced by another assessment. Late proposals and reports lose 20% per day and are not accepted after three days. The approved course outline governs assessment and absence policies.

Teamwork is evidenced by the work plan, peer assessment and bench observations. The station sheet defines the converter function, hardware and bench demonstration. These rubrics apply to all groups; additional tests are not required for full marks.